

Week 2 troubleshooting

Walk the room with this page. Most failures are declaration syntax, missing `go.mod`, or forgetting `*` on flags.

What they see	Likely cause	What you do
syntax error: non-declaration statement outside function body	<code>:=</code> at package level	<code>var name = "Ada"</code> at the top of the file
no new variables on left side of <code>:=</code>	name already exists	use <code>=</code>
undefined: name	they used <code>=</code> before <code>var / :=</code>	declare first
invalid operation: mismatched types int and float64	implicit conversion habit from JS/Python/Java	<code>float64(n)</code> or <code>int(f)</code>
cannot use x (variable of type int64) as int	<code>int</code> $\neq$ <code>int64</code>	convert explicitly
prints <code>0xc000010230</code>	they printed the flag pointer	<code>*first</code> not <code>first</code>
undefined: flag	missing import	<code>import "flag"</code>
flag provided but not defined	they used <code>-name</code> but declared <code>-first</code>	match the flag names
go: cannot find main module; see 'go help modules'	<code>go test</code> with no <code>go.mod</code>	<code>go mod init epiclearn/week02</code> in that folder
package names is not in std	they ran <code>go run names.go</code> from the wrong place	<code>cd</code> into the <code>names</code> folder; <code>go test</code>
undefined: FullName in the test	different packages, or file not saved	both files <code>package names</code> in the same folder
<code>len("café")</code> is 5	they think <code>len</code> counts letters	bytes vs runes; <code>utf8.RuneCountInString</code>

What they see	Likely cause	What you do
<code>a++</code> error inside <code>Println</code>	<code>++</code> is a statement	<code>a++</code> on its own line
<code>time: parsing time ...</code> <code>cannot parse</code>	they used <code>YYYY-MM-DD</code> as the layout	layout is <code>2006-01-02</code>
<code>go version</code> still not 1.25	leftover from week 1	use the week 1 install sheet; do not start types on apt Go
Windows: flags work in PowerShell, fail in Ubuntu	they are in the wrong terminal	stay in Ubuntu (WSL)
Binary <code>zeros</code> / <code>greet</code> on GitHub	<code>.gitignore</code> missing	ignore the binaries; <code>git rm --cached</code>

`go test` ritual (not a modules lecture)

```
cd ~/epic-go/week02/names
go mod init epiclearn/week02 # once
go test
```

If they already have a `go.mod`, do not run `go mod init` again.

Nuclear rewrite of the demo files

If a student's folder is a mess, they start a clean `~/epic-go/week02` and copy only what they typed today. Do not paste the starter until they have tried.